

Online Identification of Tripped Line for Transient Stability Assessment

Pratyasa Bhui and Nilanjan Senroy, *Member, IEEE*

Abstract—Identification of post-fault topology is critical to not only cascaded outage prevention but also prediction of post-fault dynamic behavior. Existing methods for identifying the tripped lines rely on steady state SCADA/PMU measurements, and therefore fail to accomplish their task within fractions of a second. This limits their application to online transient stability assessment. This paper presents a method for online identification of post-fault topology on the basis of information of the pre-fault operating conditions and synchronized measurements at all generator buses. The proposed technique identifies the tripped line with negligible delay, before the system reaches steady state. An innovative index, which represents the normalized kinetic energy gained by generators immediately after the fault, is used to minimize the search set of possible tripped lines. For every candidate line, the generator output powers are calculated considering the line as tripped, and the minimum error between the measured and the calculated generator powers is used to finally identify the tripped line. The IEEE 39-bus system and a 246-bus North Indian Grid representation are used to show the effectiveness of the method.

Index Terms—Fault location, fault section, line outage, post-fault topology, system topology.

I. INTRODUCTION

THE knowledge of post-fault topology is very important in predicting post-fault dynamic behavior of the system and examining steady state security of the system. The predicted post-fault dynamic behavior can be used in corrective actions like load shedding, situational awareness, generator rescheduling and restoration [1], [2]. As the transient stability after a disturbance depends on post-fault topology, knowledge of this can be used for predicting the stability after any disturbance. Four out of six major North American blackouts could have been avoided with better situational awareness of the post-fault network [3]. Key aspects of situational awareness are detection and identification of line outage and tracking of cascaded line outages.

Various methods have been proposed in the recent past to detect the correct topology and they do so in different timeframes. Most of the methods rely on SCADA data and breaker- and relay-status, which may take more than a minute to process. It

was reported in [1] that stabilizing control actions may be more effective in stabilizing an unstable event if started within 0.4 s after detecting instability or 0.8 s after fault clearing. Therefore line outages need to be detected and identified within fractions of a second for predicting transient stability after a disturbance and to further use in stabilizing controls. Most SCADA based methods for detecting topology change rely on state estimation [8]–[12]. As state estimation is done normally with 5–15 min interval, these methods are suitable for static security studies, not for stability prediction.

In this context, synchrophasors are gaining popularity in real time visualization of power system. With a PMU at every bus, the topology change can be detected and identified easily in real time. But this is unlikely to happen in the near future due to various technical and economical challenges. Several papers have been published in the last decade for predicting the stability using decision trees considering PMUs at all the generator buses [4]–[7]. As generator buses constitute a small fraction of the total number of buses, installing PMUs to make the generator buses observable is an economically viable option for disturbance monitoring and stability prediction purposes. In this work the synchrophasor data of only the generator buses has been used.

Synchrophasors have been used in [13]–[23] to quickly identify the tripped line. The Markov dependency graph model was used to locate the fault, assuming the bus phasor angles are conditionally independent and further that synchrophasor measurements cover the entire network [13]. Zhu and Giannakis used overcomplete representations to cast line outages as a sparse vector estimation problem that requires prior knowledge of system parameters [14]. Other techniques based on ambiguity group theory [15], cross entropy based optimization [16], statistical quickest change detection [17], and support vector machines [18] have also been proposed. The post disturbance steady state bus voltage angles have been used to identify single line [19] and double line outages [20]. These methods use post disturbance steady state or low pass filtered voltage, power or angle data. Again as the system takes a few seconds to come to the steady state or to filter out the oscillations effectively, these methods are suitable to study the impact of line outage on other equipments in steady state but not useful to predict the transient stability. Identifying the tripped line after a fault within fractions of a second still remains an unsolved problem.

A method to identify tripped line would be acceptable if

- it requires minimum number of PMU measurements;
- low computational burden to enables real time application;
- minimal impact of erroneous identification on dynamic behavior prediction;

Manuscript received December 09, 2014; revised February 06, 2015 and April 17, 2015; accepted May 29, 2015. This work was supported in part by the Department of Science and Technology, Government of India, under its grant DST/RCUK/SEGES/2012/02. Paper no. TPWRS-01676-2014.

The authors are with the Department of Electrical Engineering, Indian Institute of Technology Delhi, New Delhi, India (e-mail: pratyasansec@gmail.com; nsenroy@ee.iitd.ac.in).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TPWRS.2015.2440562

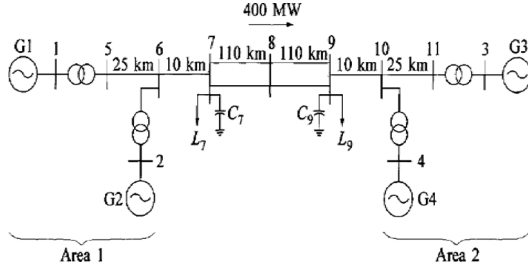


Fig. 1. Kundur's 2-Area System.

- training data base should be easily generated;
- robustness to changes in operating conditions.

In this work, a method has been proposed to identify the tripped line following a fault, fast enough for predicting stability and for use in stabilizing control. An offline database is created consisting of the kinetic energy (KE) gained by all generators during the first 0.1 s of a fault at all buses one at a time. In real time following a new fault, the KE gained by the generators is compared with the stored data to locate the candidate buses nearer to the fault location, i.e., approximate fault locations. Then nearby lines are ranked as per their probability of being the tripped line. Subsequently, the output power of generators is calculated considering each of these candidate lines as tripped one at a time. The error between the calculated and the measured power should be ideally zero when the line assumed as tripped during calculation of output power is same as the actual tripped line. Thus the error between the calculated and the measured power is used as an indicator of the tripped line.

The remainder of the paper is organized as follows: Section II discusses the proposed method. Sections III and IV present the results obtained for two test systems and comparison with other methods. Section V is the concluding section.

II. PROPOSED METHOD

The proposed method has been initially explained in the context of Kundur's 2-Area System [26] as shown in Fig. 1.

The kinetic energy gained by the generators during a fault depends on the location, duration and types of the fault. Let denote the set of actual kinetic energy gained by G generators during a fault

$$KE = [ke_1 \ ke_2 \ \dots \ ke_i \ \dots \ ke_G] \quad (1)$$

where $ke_i (= 0.5 M_i \Delta \omega_i^2)$ is the kinetic energy gained by the i th generator, M_i is the moment of inertia, $\Delta \omega_i$ is speed deviation of the i th generator.

The ke gained by a generator during the first 0.1 s is almost zero for a fault at a location which is far away from that generator and it will not change much and remain almost zero for slight changes in fault location. Therefore the ke gained by that generator will not be an efficient indicator of approximate fault location when the fault is *distant* from that generator. Whereas the ke gained by a generator is much higher and sensitive to the fault location when the fault is on nearby lines. Thus the kinetic energy gained by a generator will be an efficient indicator for faults on *nearby* lines. The Euclidean distance between the measured ke and the stored ke corresponding to a bus is used as

TABLE I
 $\overline{KE}$ FOR BALANCED FAULT AT BUS 7 OF DIFFERENT DURATIONS

Fault duration	Normalized KE			
0.05 s	$\overline{KE}_7 = [0.4021$	0.5336	0.0293	0.0350]
0.1 s	$\overline{KE}_7 = [0.4074$	0.5204	0.0340	0.0383]
0.15 s	$\overline{KE}_7 = [0.4102$	0.5121	0.0372	0.0405]

TABLE II
 $\overline{KE}$ FOR DIFFERENT TYPES OF FAULT AT BUS 7

Fault types	Normalized KE			
Balanced fault	$\overline{KE}_7 = [0.4074$	0.5204	0.0340	0.0383]
L-G fault	$\overline{KE}_7 = [0.4154$	0.4954	0.0416	0.0476]
L-L fault	$\overline{KE}_7 = [0.4181$	0.4994	0.0380	0.0445]
L-L-G fault	$\overline{KE}_7 = [0.4186$	0.5004	0.0376	0.0434]

an indicator of distance between the bus and the fault location and the buses are ranked on the basis of this distance. Therefore more emphasis is put on the ke gained by the generators nearer to the fault automatically as they are of much higher magnitude compared to the ke_i of other generators. Because of this advantage, ke outperforms fault distance, one of the popular fault location indicators, obtained from distance relays.

The normalized kinetic energy is the individual generator ke_i divided by the sum total of ke_i of all generators:

$$\begin{aligned} \overline{KE} &= [\overline{ke}_1 \ \overline{ke}_2 \ \dots \ \overline{ke}_i \ \dots \ \overline{ke}_G] \\ &= \frac{[ke_1 \ ke_2 \ \dots \ ke_i \ \dots \ ke_G]}{\sum_{i=1}^G ke_i} \end{aligned}$$

$$\text{Thus, } \sum_{i=1}^G \overline{ke}_i = 1. \quad (2)$$

The $\overline{KE}$ obtained for a fault at bus number “b” will be denoted as $\overline{KE}_b$. The $\overline{KE}$ obtained for a fault on line connecting bus “b1” and “b2” will be denoted as $\overline{KE}_{b1-b2}$. For a balanced fault of duration 0.1 s at bus 7 in Kundur's 2-Area System, the $\overline{KE}$ comes out to be

$$\overline{KE}_7 = [0.4074 \ 0.5204 \ 0.0340 \ 0.0383].$$

It has been seen that $\overline{KE}$ remains almost same for a fault location irrespective of type and duration of a fault. As fault duration for both measured and stored $\overline{KE}$ is 0.1 s, it is not compulsory for $\overline{KE}$ to remain same for different fault durations. However, if fault starting instant is not detected accurately, it can assure that $\overline{KE}$ measured in duration slightly different from 0.1 s can be compared with stored data which was calculated for a duration of 0.1 s. As the expected error in detection of fault starting instant is expected to be small, $\overline{KE}$ will not change much with that error. $\overline{KE}$ has been calculated for a balanced fault of different durations at bus 7 in Kundur's 2-Area System which has been shown in Table I. The $\overline{KE}$, for different types of fault of duration 0.1 s at bus 7 in Kundur's 2-Area System, is shown in Table II.

From Tables I and II, it is seen that the $\overline{KE}$ remains almost the same for a fault at a particular location irrespective of fault duration and type. Therefore, the $\overline{KE}$ data measured after all types

of fault can be compared with the stored $\overline{KE}$ data obtained from offline simulation for a particular fault to locate buses nearer to the fault. This makes it possible to generate data set in offline in lesser time and which is very important for updating data set after a topology change. A detailed discussion on the benefit of using $\overline{KE}$ is included in the Appendix.

Using generator buses' PMU data, the pre-fault topology and near real time loading condition, the output power of generators is calculated considering one line outage at a time. A line will be considered as tripped if the error between the calculated power considering the line as tripped and the measured power at the generator buses is minimum. The proposed method has four steps:

A. Step 1: Data Set Generation in Offline

Using near real time loading condition and topology, the $\overline{KE}$ gained during first 0.1 s of fault at a bus "b" is calculated from simulation and stored:

$$\overline{KE}_b = [\overline{ke}_1 \ \overline{ke}_2 \ \dots \ \overline{ke}_i \ \dots \ \overline{ke}_G]. \quad (3)$$

This is repeated for every bus "b". For each bus "b" of the system, there will be a stored vector $\overline{KE}_b$ of length equal to the total number of generators. After a fault, this data set will be used to locate the buses and lines closer to the fault.

B. Step 2: Locating Buses Closer to the Fault

After a fault, $\overline{KE}$ is calculated from speed of the generators measured by PMU which is denoted as $\overline{KE}_L$ for faults on line "L". For example, for a balanced fault at 50% of the line connecting bus 6 and 7, i.e., Line₆₋₇ in Kundur's 2-Area system, $\overline{KE}_L$, i.e., $\overline{KE}_{6-7}$ comes out to be [0.4203 0.5146 0.0304 0.0346]. For a fault on line "L", $\overline{KE}_L$ should be closer to stored normalized $\overline{KE}_b$ for buses "b" closer to the fault. Using the k^{th} Nearest Neighbor (k-NN) method [25], the Euclidean distance D_b between $\overline{KE}_L$ and $\overline{KE}_b$ for each bus "b" is calculated. A small D_b indicates that the fault is nearer to the bus "b". The buses are ranked in ascending order of D_b .

C. Step 3: Ranking of the Lines Closer to the Fault

In this step an index D_L is calculated for every line "L" which is the sum of the Euclidean distance D_b , obtained in step 2, for the two buses connected by Line "L". The lower the index, the higher is the possibility of the line of being the faulted line. Lines are ranked in ascending order of D_L . If line "L" connects buses "b₁" and "b₂", then D_L equals to ($D_{b1} + D_{b2}$).

D. Step 4: Identification of Exact Tripped Line

A line will be assumed to be the tripped line at a time and the output power of the generators will be calculated. Calculated power assuming line "L" as tripped should match the measured power if line "L" is the actual tripped line. It will be repeated for all the lines. Loading condition can be obtained from SCADA data at an interval of 2–5 s. As, loads in power system change very slowly, the post-fault loading condition can be approximated as the pre-fault loading condition already available from SCADA and recalculated using a dynamic load model. The base case topology can be obtained from the latest state estimation. With the base case topology and the post-fault loading condition

known, the post-fault output power of the generators can be calculated from the post-fault voltage and angle data of generator buses obtained from PMU as follows. If loads are considered as admittance load, the post-fault power can be calculated using reduced admittance bus matrix [29]. Considering net zero current injection at non-generator buses, the current injection at all buses can be represented as

$$\begin{pmatrix} I_g \\ 0 \end{pmatrix} = \begin{pmatrix} Y_{gg} & Y_{gr} \\ Y_{rg} & Y_{rr} \end{pmatrix} \begin{pmatrix} V_g \\ V_r \end{pmatrix} \quad (4)$$

where the subscript g denotes generator bus and subscript r denotes the remaining buses.

Eliminating non-generator buses

$$I_g = [Y_{gg} - Y_{gr}Y_{rr}^{-1}Y_{rg}] V_g = Y_{red}V_g. \quad (5)$$

The voltage vector V_g can be obtained from the voltage magnitude and the angle data measured by the PMU. Using the vector I_g and V_g , output power at generator buses can be calculated as

$$P_{cal} = P_g = V_g I_g. \quad (6)$$

Let the pre-fault calculated and measured power are

$$P_{cal} = [P_{cal1} \ P_{cal1} \dots P_{cali} \dots P_{calG}] \quad (7)$$

$$\text{and } P_{meas} = [P_{meas1} \ P_{meas2} \dots P_{measi} \dots P_{measG}]. \quad (8)$$

If the line "L" is considered to be the tripped line, network topology and therefore Y_{red} are recalculated. The post-fault output power considering line "L" as tripped is calculated from (6) as

$$P_{calL} = [P_{cal1L} \ P_{cal2L} \dots P_{caliL} \dots P_{calGL}]. \quad (9)$$

The change in the calculated power from the pre-fault to the post-fault state, i.e., ΔP_{calL} is taken instead of the output power P_{calL} and compared with change in measured power from pre-fault to the post-fault.

The change in the output power from pre-fault to post-fault condition, considering line "L" as tripped, is denoted as

$$\Delta P_{calL} = [\Delta P_{cal1L} \ \Delta P_{cal2L} \dots \Delta P_{caliL} \dots \Delta P_{calGL}]. \quad (10)$$

Similarly the change in the measured generators power from pre-fault to post-fault condition can be denoted as

$$\Delta P_{meas} = [\Delta P_{meas1} \dots \Delta P_{measi} \dots \Delta P_{measG}]. \quad (11)$$

Thus the norm of the difference between the change of the measured and the calculated power, assuming line "L" as tripped, is

$$P_{errL} = |\Delta P_{meas} - \Delta P_{calL}|. \quad (12)$$

As ΔP_{meas} is the difference between the pre-fault and the post-fault actual measured power and ΔP_{calL} is calculated considering line "L" as tripped, ΔP_{meas} will be same for all lines whereas ΔP_{calL} will be different for different lines. If line "L" is the actual tripped line, then P_{errL} calculated considering line "L" as tripped should be close to zero. If P_{errL} is calculated for all lines, then the line that gives minimum P_{errL} can be considered as the tripped line. If number of lines is N_L , then P_{errL} is

to be calculated N_L times which is not acceptable because of the computation time requirement. Therefore the initial ranking of the lines on the basis of D_L can be used to reduce the number of lines for which $Perr_L$ is calculated. As the application of this technique is in real time transient stability assessment, the tripped line has to be identified very quickly. This will be possible only if the initial ranking of the lines on the basis of D_L is good enough such that the actual tripped line ranks within first few lines.

III. RESULTS

The proposed method has been tested on different test systems. As loads may vary from the operating condition for which data set was generated offline and stored, the proposed method was tested by varying the loads randomly within a certain margin. The performance with load variation along the daily load curve would determine the interval at which data set has to be generated. The average number of lines tested indicates the computational burden of the proposed method.

A. Kundur's 2-Area System

Large number of case studies was done on this Test System, out of which only two have been discussed in detail. Each of the 6 non-transformer branches was replaced with a double circuit line with same equivalent impedance and only one circuit was tripped after fault.

1) *Line 6-7 Outage:* $\overline{KE}$ gained by the generators was calculated for a balanced fault of duration 0.1 s at a bus "b" and then $\overline{KE}_b$ gained by the generators in first 0.1 s was stored. This was repeated for every bus. Then a balanced fault of 0.1 s duration was created on line 6-7 at 50% of the line and $\overline{KE}_{6-7}$ was calculated. Using *kth Nearest Neighbor (k-NN)* method, the Euclidean distance D_b was calculated from $\overline{KE}_{6-7}$ to $\overline{KE}_b$ for each bus "b". For each line "L" connecting buses b_1 and b_2 , an index D_L was calculated which the sum of D_{b1} and D_{b2} was. The lines were ranked in ascending order of D_L and the difference between the change in measured and the calculated power $Perr_L$ was calculated for outage of every line "L". It is seen from Table III that $Perr_L$ calculated considering line 6-7 as tripped is minimum.

After $Perr_L$ comes below a pre-determined threshold, no further line outage is to be tested. If the threshold for $Perr_L$ was taken as 0.05, after testing the first line, i.e., Line₆₋₇ only, $Perr_L$ was obtained below threshold and the Line₆₋₇ was considered as the tripped line. Faults may be self clearing and no line may be tripped. Therefore $Perr_L$ was calculated for no line outage also.

2) *Line 7-8 Outage:* Similar to the procedure followed in Line₆₋₇ outage, $Perr_L$ was calculated considering one line as tripped at a time. The result is shown in Table IV. In this case, the $Perr_L$ becomes below threshold after two line outage calculations and after that no further line outage needs to be tested. Therefore the number of lines tested is two in this case. It is seen that $Perr_L$ becomes minimum when Line₇₋₈ is considered as tripped. But $Perr_L$ calculated with Line₈₋₉ as tripped also show very low value. This is because the impacts of Line₇₋₈ outage and Line₈₋₉ outage on system dynamics are similar.

TABLE III
 $Perr_L$ FOR ALL LINE OUTAGES FOR FAULT ON Line₆₋₇

bus	7	6	2	8	5	1	No
D_b	0.0145	0.0257	0.0993	0.305	0.4225	0.57	Outage
Line	6-7	7-8	5-6	8-9	9-10	10-11	-
D_L	0.0402	0.3195	0.448	1.021	1.455	1.6	-
$Perr_L$	0.0034	0.3957	2.4576	0.3675	1.9888	2.4597	1.0656

TABLE IV
 $Perr_L$ FOR ALL LINE OUTAGES FOR FAULT ON Line₇₋₈

bus	7	6	2	8	5	9	No
D_b	0.1123	0.1425	0.1463	0.1879	0.487	0.603	Outage
Line	6-7	7-8	5-6	8-9	9-10	10-11	-
D_L	0.2549	0.3002	0.6296	0.7609	1.2301	1.386	-
$Perr_L$	0.3447	0.0296	2.8408	0.0608	2.2685	2.5461	1.1795

TABLE V
ACCURACY AND AVERAGE COMPUTATIONAL BURDEN FOR IDENTIFYING
TRIPPED LINE IN KUNDUR'S 2-AREA SYSTEM

Loading Condition Variation	Fault distance as % of line			
	Wrong identification	Average number of lines tested	Wrong identification	Average number of lines tested
0%	0/6	1.5	0/6	1.33
3% randomly	0/6	1.33	0/6	1.33
5% randomly	0/6	1.5	0/6	1.33
20% randomly	0/6	1.5	0/6	1.5
3% uniformly	0/6	1.33	0/6	1.33
5% uniformly	0/6	1.33	0/6	1.33
10% uniformly	0/6	1.33	0/6	1.33
20% uniformly	0/6	1.33	0/6	1.33

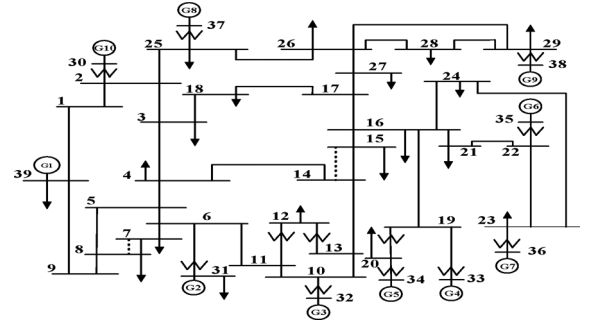


Fig. 2. Network diagram of IEEE 39-Bus test system.

The proposed method has been tested for different loading conditions and at different distance of the lines. Results are given in Table V. The average number of lines tested is also small enough which indicates that this method can detect tripped line with small computational burden. This is very important for stabilizing control.

B. IEEE 39-Bus Test System

Like Kundur's 2-Area system, each line of the 34 non-transformer branches of IEEE 39-Bus test system [27] was replaced with a double circuit line with same equivalent impedance. The fault was created on a non-transformer branch and only one circuit was tripped after the fault. The total number of the non-transformer branches is 34. The test system has been shown in Fig. 2. The results are summarized in Table VI.

TABLE VI
ACCURACY AND AVERAGE COMPUTATIONAL BURDEN FOR IDENTIFYING THE
TRIPPED LINE IN IEEE 39-BUS SYSTEM

Loading Condition Variation	Fault distance as % of line			
	50%		75%	
	Wrong identification	Average number of lines tested	Wrong identification	Average number of lines tested
0%	0/34	1.5882	0/34	1.7647
3% randomly	0/34	1.6176	0/34	1.7353
5% randomly	0/34	1.6471	0/34	1.7647
20% randomly	0/34	1.7647	0/34	1.8235
3% uniformly	0/34	1.5588	0/34	1.7647
5% uniformly	0/34	1.5882	0/34	1.8235
10% uniformly	0/34	1.7647	0/34	1.9412
20% uniformly	0/34	1.8529	0/34	1.9706

TABLE VII
ACCURACY AND AVERAGE COMPUTATIONAL BURDEN FOR IDENTIFYING
TRIPPED LINE IN NORTH INDIAN GRID

Loading Condition Variation	Fault distance as % of line			
	50%		75%	
	Wrong identification	Average number of lines tested	Wrong identification	Average number of lines tested
0%	0/372	3.0403	0/372	3.0242
3% randomly	1/372	3.0645	1/372	3.0376
5% randomly	2/372	3.0914	2/372	3.1048
20% randomly	2/372	3.1720	2/372	3.1452
3% uniformly	1/372	3.0591	1/372	3.0323
5% uniformly	0/372	3.0699	0/372	3.0833
10% uniformly	1/372	3.1129	1/372	3.0887
20% uniformly	2/372	3.1989	2/372	3.1855

C. North Indian Grid

Proposed method has also been tested on 57-generator, 246-bus North Indian Grid [28] for different loading conditions and fault locations on 372 lines of the system. Actually the system had 60 generators, but generators at buses 40 and 42 were kept out of service. Unlike the two systems tested above, in this system no double circuit line was created replacing a single circuit line. Faults were cleared by opening the faulted line. The detailed result as tabulated in Table VII, confirms its capability to identify the tripped line with high accuracy and relatively small computational burden for a large grid.

D. Impact of Wrong Identification in Predicting Dynamic Behavior

It is possible that the method fails to identify the tripped line and shows a healthy line as the tripped line. This happens when P_{errL} is smaller than a preselected threshold value for the outage of a line other than the actual tripped line. It is very important that the control actions taken on the basis of this wrongly identified tripped line works effectively. This is possible only if the post disturbance dynamic behavior of the system is similar for the outage of the actually tripped line and the wrongly identified line which shows P_{errL} below threshold. Therefore the correlation between P_{errL} and error in predicting post-fault dynamic behavior needs to be studied.

Suppose “ L_1 ” is the actual tripped line. P_{errL} calculated considering line “ L_1 ” as tripped should be minimum. If line “ L_2 ” ranks above line “ L_1 ” in the ranking done using D_L and

P_{errL_2} is below threshold value, then line “ L_2 ” will be wrongly considered as the tripped line:

$$\begin{aligned} P_{errL_1} &= |\Delta P_{meas} - \Delta P_{calL_1}| \\ P_{errL_2} &= |\Delta P_{meas} - \Delta P_{calL_2}|. \end{aligned} \quad (13)$$

As P_{errL_1} and P_{errL_2} both are below threshold and ΔP_{meas} is same for both the cases

$$P_{errL_1} - P_{errL_2} < threshold. \quad (14)$$

$$\text{From (13), } \Delta P_{calL_1} - \Delta P_{calL_2} < threshold. \quad (15)$$

As the pre-fault calculated power for healthy network is same for both the lines, the difference between the post-fault power for outage of lines “ L_1 ” and “ L_2 ” should be below threshold:

$$P_{calL_1} - P_{calL_2} < threshold. \quad (16)$$

Considering constant mechanical power input to the generators during the period of observations, the post-fault accelerating power for outage of line “ L_1 ” and “ L_2 ” can be written as

$$\begin{aligned} P_{aL_1} &= P_{mech} - P_{calL_1} = \Delta P_{calL_1} \\ P_{aL_2} &= P_{mech} - P_{calL_2} = \Delta P_{calL_2}. \end{aligned} \quad (17)$$

From (15) and (17), it can be concluded that as ΔP_{calL_1} and ΔP_{calL_2} are nearly same, post-fault accelerating power considering line “ L_1 ” outage and line “ L_2 ” outage are also nearly same. According to the swing equation, angle and speed behavior depends on accelerating power for each generators. Thus the post-fault dynamics would also be similar for line “ L_1 ” outage and line “ L_2 ” outage for the same fault.

This infers that the post-fault dynamic behavior of the system considering line “ L_1 ” outage and line “ L_2 ” outage for the same fault will be similar if P_{errL_1} and P_{errL_2} are below threshold. If the proposed method wrongly identifies a line as tripped line which shows low value of P_{errL} , predicted dynamic behavior will be similar to the actual dynamic behavior. This has been verified with two examples:

1) *Kundur's 2-Area System*: A balanced fault was simulated at bus 8 (Fig. 1) at 1 s followed by the line 7–8 tripping. The P_{errL} was calculated for different line outages as: 0.0302, 0.0614, 0.3342, 0.79, 2.1913, and 2.5038 for line 7–8, 8–9, 6–7, 5–6, 9–10, and 10–11, respectively. As mentioned earlier, each line in the original system was replaced with a double circuit line of which one was tripped to clear the fault.

If the P_{errL} for the outage of a healthy line is small, then the error in predicting generator swings considering that line as tripped will also be small. In Fig. 3 a balanced fault was created at bus 8 at 1 s and cleared at 1.1 s by tripping line 7–8. The swing curves of two generators (in CoI reference) obtained from this event were compared with other events in which different other lines were tripped at 1.1 s instead of line 7–8. It can be seen that the swing curve for line 8–9 outage is the closest curve to the swing curve for outage of the line 7–8. This agrees with the values of P_{errL} obtained for both lines 8–9 and 7–8.

2) *IEEE 39-Bus Test System*: A balanced fault near bus 5 on line_{5–6} was created in the IEEE 39-Bus test system and cleared by tripping Line_{5–6}. The network solution considering

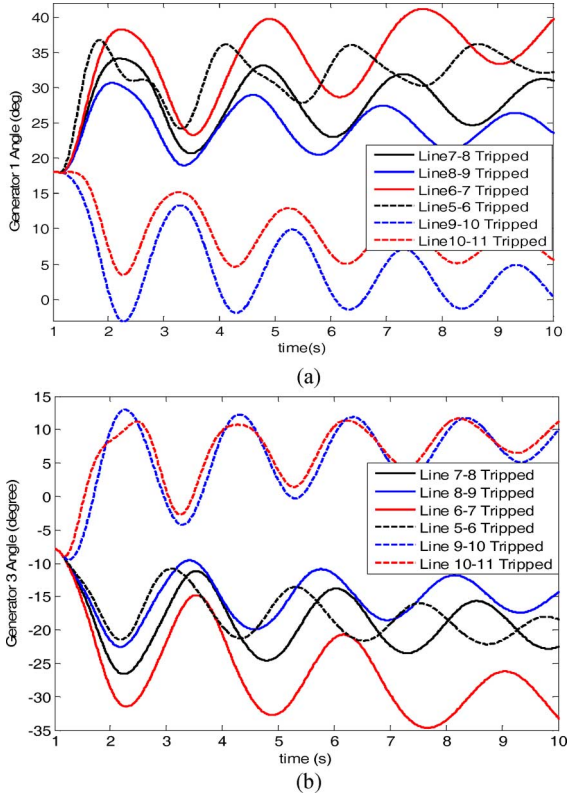


Fig. 3. Swing curves for a 0.1 s fault near bus 8 and different line tripping at 1.1 s. (a) Generator 1 angle. (b) Generator 3 angle.

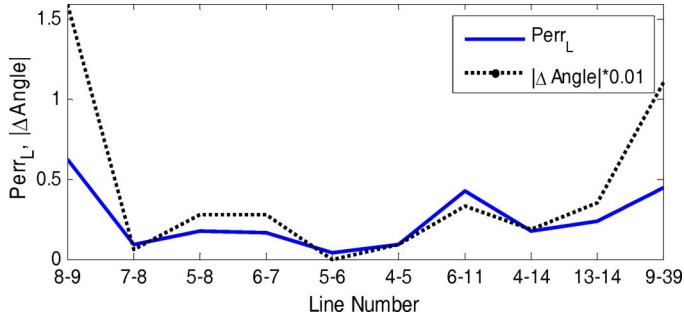


Fig. 4. Error in predicting dynamic behavior of post-fault system and $Perr_L$ considering line “L” as tripped.

Line₅₋₆ outage gives minimum $Perr_L$. But Line₄₋₅ and Line₇₋₈ also show very low value of $Perr_L$. As shown in the previous paragraph, post-fault swing curves can be obtained for outages of different lines. Suppose swing curves for outage of line₄₋₅ is subtracted from the swing curve for outage of line₅₋₆ and then the difference is stored in a matrix ΔAngle . Norm of that matrix will indicate the error in predicting the generator angles if line₄₋₅ is assumed as the tripped line instead of the actual tripped line, i.e., line₅₋₆. $Perr_L$ and $|\Delta\text{Angle}|_L$ were calculated for every line “L” for a fault on line₅₋₆ near bus 5. Fig. 4 shows that $|\Delta\text{Angle}|_L$ is small when $Perr_L$ considering line “L” as tripped is also small. Therefore even if this method wrongly identifies a line other than the actual tripped line, predicted generator angles will be similar to the actual angles.

E. Line Outage Causing Islanding in the Grid

Recently published methods [8]–[10] fail to identify the tripped line if the post disturbance system forms islands. The lines that cause islanding in the system, are very crucial from frequency stability and blackout protection point of view. Therefore the performance of the method in such situation also needs to be checked.

1) *Line 7–8 Outage in Kundur's 2-Area System:* A balanced fault of 0.1 s duration is created on line₇₋₈ at 50% of the line and then cleared by tripping both circuits of the double circuit line₇₋₈. It separates the system into two areas: area₁ consists of buses 1, 2, 5, 6, and 7 and area₂ consists of buses 3, 4, 8, 9, 10, and 11. First, the lines are ranked using index D_L and then $Perr_L$ is calculated considering one line “L” as tripped at a time. Results are given in Table VIII. It is seen from Table VIII that $Perr_L$ is minimum when it calculated assuming line₇₋₈ as tripped and the number of line outage tested is two which is reasonably small.

IV. COMPARISON

Recently published papers, to identify the tripped lines using PMU data, were mainly focused at steady state security. The proposed method uses PMUs located at generator buses to identify the tripped line just after its tripping, with negligible delay. The performance of the proposed method has been compared with that of the LIS method [15].

Tripped line can only be identified only if the line is tripped because of faults which is the most common case. It can be seen from Table IX that proposed method performs better for line outage due to faults with negligible delay. Therefore it is most suitable for transient stability studies.

V. ISSUES WITH PRACTICAL IMPLEMENTATION

A. Error in Load Modeling

A key step in the proposed technique is the calculation of generator power output and matching it with measured values obtained using PMUs. Any time domain simulation first calculates the angles and voltages of generators by solving differential equations. On the basis of these calculations the output power of generators are calculated by solving algebraic equations using the existing topology and loads. Similarly, the proposed method in this paper also calculates the generator power using the angles and voltages obtained from synchrophasor measurements. Therefore, calculating the generator power with nonlinear load models is well established and frequently used in time domain simulations (solution of algebraic equations) [33, Section 6.3].

In practice, it may not be possible to always match the calculated values with the measurements. It was reported in [15] that the accuracy of line outage identification drastically reduces with changes in pre-fault to post-fault loading conditions. As the analysis was done with steady state data in [15], it was assumed that pre-fault and post-fault loading were equal. Thus, with PMU coverage of 50% of the buses, the accuracy reduced from 86.67% with no change in loading condition to 68.83% for 1% change, 56.91% for 2% change and 40.51% for 5% change in loading condition. In this work, the post-fault load should be

TABLE VIII
 $Perr_L$ FOR ALL LINE OUTAGE CALCULATIONS FOR FAULT ON Line₇₋₈

bus	7	2	6	8	5	9
D_b	0.0947	0.1266	0.1274	0.2079	0.485	0.623
Line	6-7	7-8	5-6	8-9	9-10	10-11
D_L	0.222	0.3026	0.6123	0.8308	1.2697	1.4228
$Perr_L$	7.134	0.081	9.14	0.16	11.62	7.91

TABLE IX
 COMPARISON BETWEEN PROPOSED METHOD AND LIS METHOD

	LIS [15]	Proposed Method
Accuracy	[PMU coverage=30%] IEEE 14 bus system: 80% IEEE 57 bus system: 65% IEEE 118 bus system: 75%	Kundur's 2 Area system: PMU coverage: 36% Accuracy: 100% IEEE 39 bus System: PMU coverage: 25% Accuracy: 100% 246 bus NR Grid: PMU coverage: 16% Accuracy: 99.7%
Time required after fault clearing (including data transmission delay):	In range of 5-60 second, as it requires steady state/LP Filtered data	<0.2 second
Post fault loads	Assumed equal to pre-fault values	Depends on load model
Does it work if the post fault system is islanded?	No	Yes
Causes behind line outage	Independent of line-outage cause	Only fault induced line-outages
Suitable applications	Steady state security studies	Transient stability and steady state security

calculated from generator bus voltage and angle data, the dynamic load models and latest loading conditions obtained from SCADA data.

Some error may also be introduced due to error in load modeling and measurement noise. It is almost impossible to model dynamic load behavior accurately. In [31], different representative load models were compared with actual measured data. The error between measured and calculated real power was 0.56% for a third order induction machine load model, 0.62% for a first order induction machine load model, 0.7% for an exponential load model and 0.79% for the polynomial model. The performance of the proposed method was tested with different levels of error in the calculated generator power. This was accomplished by varying the load models during simulation, while using the admittance representation to calculate the power. It was seen that although the $Perr_L$ increases with error in calculating the active power, the line ranking however remains unaffected. From offline studies, the maximum rank of the faulted line R_m for fault on any line in the system may be known in advance. As $Perr_L$ is sensitive to error in calculating output power of the generators, it is therefore advised to test R_m number of lines and the line that gives minimum $Perr_L$ will indicate the tripped line.

In the IEEE 39-Bus test system, for every fault the first four lines from the ranking were tested. The results obtained with different error between output power obtained from measurement (represented by simulation data here) and calculation using generator bus data for IEEE 39-Bus system has been shown in

TABLE X
 IMPACT OF ERROR IN CALCULATING OUTPUT POWER

Percentage error between measured and calculated power of generators	Wrong identification
No error	0/34
0.5%	0/34
0.75%	0/34
1.1%	1/34
2.2%	2/34
2.5%	2/34

Table X. The accuracy of the proposed method decreases with increase in error in calculating the output power of generators. Therefore, accurate dynamic load models are recommended for best results. This finding is also supported by earlier works [15].

B. Multiple Line Outages

Multiple line outages at a time may be classified into two categories: 1) independent, and 2) interlinked outages. The probability of multiple independent line outages occurring together (the time span of interest here is less than a second) is relatively low and therefore not considered here. Interlinked outages due to a single fault/disturbance may occur because of zone-3 operation of relay, breaker failure or overloading of other lines. Typical intentional time delay is 0 s for zone 1 operation, 0.3–0.6 s for zone 2, 1.8 s for zone 3 and 0.15–0.2 s for breaker failure protection [34]. Zone-1 and zone-2 operations of relays normally don't trip multiple lines, but zone-3 and breaker failure may trip multiple lines feeding the fault current. Relay and breaker operation time have to be added with the intentional delay. Moreover, the outage of a line may cause overloading in other equipment like transformer, lines, etc. that further cause outage of parallel lines. Using line outage distribution factors (LODF), overloading in other lines may be detected instantaneously in a non-iterative manner. Each of the over loaded lines may be tested for trip status by comparing the measured and calculated power as proposed in this paper. As line outages due to overloading occurs after relatively longer time (typically minutes), the available methods based on state estimation [8]–[12] or load flow [13]–[17] may also be used for this purpose. Multiple line outages due to relay or breaker mal-operation must be identified quickly as it happens within a very short time span and can create system-wide instability.

Multiple line outage due to breaker failure is explained with the help of IEEE 39-Bus test system (zoomed portion) in Fig. 5. If the breaker CB2 fails to clear a fault on line 4–5, the fault will be cleared by opening lines 3–4 and 4–14. It can be seen that for a failure of a breaker on bus 4, all the lines connected to bus 4 have to be tripped to clear the fault. As per the proposed method in this paper, for a fault on 4–5 the buses are ranked in ascending order of distance obtained using k -NN search method. Thus, the outages of lines connected to the ranked buses may be checked. If the fault duration is less than the minimum delay setting $Perr_L$ (say 0.3 s), only single line outage may be checked. If the fault duration is more than τ_d , the single line outages as well as multiple line outages need to be tested for. Starting from the bus with lowest index D_b , the connected lines are assumed

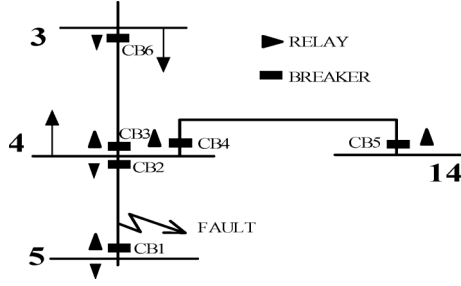


Fig. 5. Multiple line outage due to relay/breaker failure.

TABLE XI
IDENTIFICATION MULTIPLE LINE OUTAGE

bus	9	4	5	8	7	14	No
D_b	0.0525	0.055	0.068	0.069	0.0714	0.0879	Outage
Single Line Outage							
Line	8-9	4-5	5-8	7-8	4-14	5-6	-
D_L	0.1215	0.123	0.1371	0.1405	0.1429	0.1631	-
$Perr_L$	1.756	0.4164	0.4941	0.4568	0.4024	0.3885	0.5029
Multiple Line Outage							
Bus	9	4	5	8	7	14	No outage
Assumed	9-8	4-3	5-4	8-5	7-6	14-13	
tripped lines	9-39	4-5	5-6	8-7	7-8	14-15	
		4-14	5-8	8-9		14-4	
$Perr_L$	2.8	0.03	0.40	2.03	0.44	0.29	0.50

as tripped and τ_d is calculated. The minimum of $Perr_L$ out of all single and multiple lines will indicate the tripped line(s).

The results for the outage of lines 3-4, 4-14 and 4-5 due to a 0.33 s fault on line 4-5 in IEEE39-Bus test system, is shown in Table XI. Each line was replaced by a double circuit line and only one circuit was disconnected for the indicated lines.

Similar to the method explained in Section II, first single line outages are tested. It can be seen that $Perr_L$ becomes minimum when line 5-6 is assumed as tripped in single line outage cases. As the fault duration is higher than delay setting τ_d (0.3 s), multiple line outages need to be checked. Starting from the first bus in the ranking using index D_b (see first row), all the lines connected to the bus are identified. For bus 9, connected lines are line 9-8 and line 9-39. Similarly, the lines connected to bus 4 are line 4-3, line 4-5 and line 4-14. For each bus all the connected lines are assumed as tripped and $Perr_L$ calculated. For bus 9, considering line 9-8 and line 9-39 as tripped, $Perr_L$ becomes 2.7886. Similarly, for bus 4, $Perr_L$ comes out to be 0.0309. The minimum $Perr_L$ among all single and multiple line outages tested, indicates the correct tripped line(s). In this case, $Perr_L$ becomes minimum when line 4-3, line 4-5, and line 4-14 are assumed as tripped.

C. Applications of the Tripped Line Identification

1) *Static Security*: Using LODFs, limit violations of other equipment like lines, transformers, generators etc. can be checked in real time. As the time constraint to identify the tripped line is not so stringent, the time available for taking corrective action will be relatively more.

2) *Transient Stability Prediction and Control*: For a given operating condition, the transient stability of a system after a

fault depends on fault location, duration, fault resistance and post-fault topology. Impact of the first three factors is inherently included in measured kinetic energy of the generators. Therefore, if the post-fault topology is known from the knowledge of both the tripped line and pre-fault topology, the transient stability after a fault may be predicted. A detailed discussion of transient stability prediction methods is beyond the scope of this paper.

VI. CONCLUSION

This paper proposes a novel technique that can identify the tripped line(s) after a fault with negligible computational burden. It uses PMU data at generator buses only. The normalized kinetic energy, i.e., $\overline{KE}$ is calculated after a fault from the PMU data. It is compared with the $\overline{KE}$ stored for faults at each bus calculated from offline simulation to get the most likely tripped lines. Subsequently, the output powers of all generators are calculated considering every candidate line as tripped. Numerical results show that the difference between the change in the measured and the calculated power can identify the tripped line with reasonable accuracy and computational burden. This method can identify self clearing faults also for which no line is tripped as well as multiple line outages also. It has also been shown that the method is reasonably accurate with variations in loading condition and power islanding. As the computational burden is very small and the method does not require the dynamics to die out, it is capable of detecting the tripped line just after the fault clearing. The application of the proposed technique is in the development of response based stabilizing control strategy.

APPENDIX

Observation 1: Normalized kinetic energy, i.e., $\overline{KE}$ gained by the generators remains nearly same for different types of fault at a particular location.

Explanation: Active power transfer from a generator decreases (i.e., accelerating power $P_a = P_{mech} - P_{elec}$ increases) when the fault current or fault MVA supplied by the generator increases [see [32, Ch. 6]]. Therefore, the kinetic energy gained by the generators in first 0.1 s of a fault is proportional to the fault MVA supplied by the generator. Using star delta transformation, it can be shown that with reduction in the impedance of the fault current path, the equivalent impedance for power transfer to grid increases.

Fig. 6(a) shows a fault on one of two lines connecting two machines. The equivalent star-delta transformation [Fig. 6(b)-(c)] results in a Thevenin equivalent circuit at the fault point. In Fig. 6(c)-(d), the effect of negative and zero sequence components due to an unbalanced fault can be represented with a reactance X_{EF} . The value of X_{EF} depends on fault type as given in Table XII.

From Fig. 6(c), it can be said that the percentage of the total fault current supplied by a particular generator can be obtained from current division law which depends on the ratio of $(X'_d + X_T + X_A)$ and $(X_B + X_s)$. It may be inferred from Fig. 6(d) that the total fault MVA supplied by all the generators varies for different types of fault due to change in X_{EF} . As the rest of the

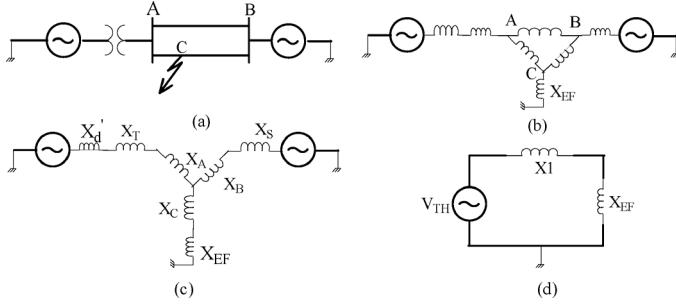


Fig. 6. (a) Line fault on one of two parallel lines connecting two machines. (b) Impedance diagram for the fault point. (c) Star-delta transformation. (d) Thevenin's equivalent representation at point of fault.

TABLE XII
EFFECTIVE FAULT IMPEDANCE FOR DIFFERENT TYPE OF FAULT

	L-L-L	L-L	L-L-G	L-G
X_{EF}	0	X_2	$X_2 X_0$	$X_0 + X_2$

network remains same irrespective of the fault, hence by current division law, the proportion of the total fault current supplied by a generator remains same for different types of fault at any given location. Therefore, after dividing with total kinetic energy, the $\overline{KE}$ remains same for all types of fault at a particular location. The equivalent circuit for calculating fault MVA has been shown in Fig. 7 for L-L-L and L-L fault. As $X_2 \cong X_1$, the total fault MVA supplied in an L-L fault is nearly *half* of the fault MVA supplied in an L-L-L fault. Thus, the accelerating power is half in an L-L fault as compared to an L-L-L fault for all generators.

Example 1: The accelerating power of individual generators for two types of fault at different buses in the Kundur's 2-Area system has been shown in Fig. 8. For faults at any bus, the accelerating power reduces to half in an L-L fault compared to an L-L-L fault. The kinetic energy gained by the generators in 0.1 s, being proportional to the accelerating power, will also show a similar trend as in Fig. 8. Further, from Table XIII, it may be concluded that the $\overline{KE}$ is almost unaffected by the fault type.

Example 2: In the three-machine system of Fig. 9, all lines and machines are identical. The line impedances are indicated in the figure, along with the transformer leakage reactance plus the generator transient reactance. The positive and negative sequence impedances are considered equal.

The reduced admittance matrix of the pre-fault system is shown in the equation at the bottom of the page. The reduced admittance matrices of the faulted system for an L-L-L and L-L fault are shown in the equation at the top of the next page.

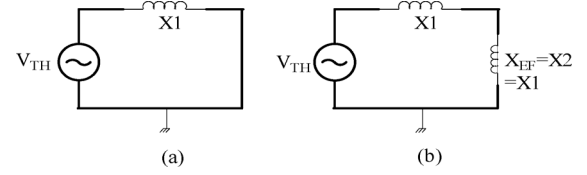


Fig. 7. Equivalent Circuit for L-L-L fault and L-L fault. (a) L-L-L Fault. (b) L-L Fault.

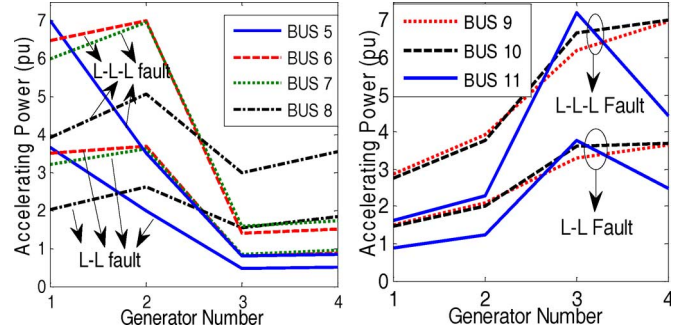


Fig. 8. Accelerating Power of generators during L-L-L and L-L fault at different buses in Kundur's 2-Area System.

TABLE XIII
NORMALIZED KE, i.e., $\overline{KE}$ FOR L-L-L AND L-L FAULTS AT DIFFERENT BUSES IN KUNDUR'S 2-AREA SYSTEM

Faulted Bus Number	Normalized Kinetic Energy i.e. $\overline{KE}$			
5	L-L-L fault: [0.7514 0.2176 0.0150 0.0160]			
	L-L fault: [0.7253 0.2362 0.0179 0.0206]			
6	L-L-L fault: [0.4425 0.5035 0.0260 0.0280]			
	L-L fault: [0.4655 0.4796 0.0300 0.0248]			
7	L-L-L fault: [0.4074 0.5204 0.0340 0.0383]			
	L-L fault: [0.4181 0.4994 0.0380 0.0445]			
8	L-L-L fault: [0.2521 0.3859 0.1584 0.2036]			
	L-L fault: [0.2513 0.3820 0.1610 0.2057]			
9	L-L-L fault: [0.0809 0.1338 0.3548 0.4305]			
	L-L fault: [0.0795 0.1339 0.3654 0.4212]			
10	L-L-L fault: [0.0711 0.1186 0.3908 0.4195]			
	L-L fault: [0.0693 0.1183 0.4065 0.4059]			
11	L-L-L fault: [0.0389 0.0666 0.6317 0.2628]			
	L-L fault: [0.0385 0.0676 0.6178 0.2761]			

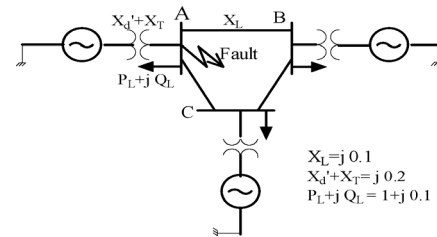


Fig. 9. Three-machine system.

$$\mathbf{Y}_{pre} = \begin{pmatrix} 0.322 - j2.9521 & 0.3018 + j1.3363 & 0.3018 + j1.3363 \\ 0.3018 + j1.3363 & 0.322 - j2.9521 & 0.3018 + j1.3363 \\ 0.3018 + j1.3363 & 0.3018 + j1.3363 & 0.322 - j2.9521 \end{pmatrix}$$

$$\mathbf{Y}_{\text{fault}_{\text{LLL}}} = \begin{pmatrix} -j5 & 0 & 0 \\ 0 & 0.0648 - j3.82 & 0.0446 + j0.4683 \\ 0 & 0.0446 + j0.4683 & 0.0648 - j3.82 \end{pmatrix}$$

$$\mathbf{Y}_{\text{fault}_{\text{LL}}} = \begin{pmatrix} 0.1671 - j3.9608 & 0.1555 + j0.6779 & 0.1555 + j0.6779 \\ 0.1555 + j0.6779 & 0.1969 - j3.3798 & 0.1766 + j0.9086 \\ 0.1555 + j0.6779 & 0.1766 + j0.9086 & 0.1969 - j3.3798 \end{pmatrix}$$

The active power output of a generator depends on the non-diagonal elements and the real part of the diagonal element. It can be seen that the “changes” in these active power related terms from their pre-fault values is half in the case of an L-L fault compared to the L-L-L fault. Therefore the drop in output power during an L-L fault is half the drop during L-L-L fault for all the generators. Thus, kinetic energy gained by the generators will also vary by the same proportion and the $\overline{KE}$ will be almost same for different types of fault at any given location.

Note:

- 1) For L-G and L-L-G faults, the $\overline{KE}$ gained will be nearly same to that of an L-L-L fault, only if X_0/X_1 ratio for all the lines are nearly the same and every generator transformer has a ground path for zero sequence current to flow. This is usually the case and hence the $\overline{KE}$ calculated for balanced fault can be used in unbalanced ground faults too. If X_0/X_1 ratio is not the same for all lines, the $\overline{KE}$ must be calculated for unbalanced ground faults separately from offline simulations and used for ranking the lines after a ground fault.
- 2) For a generator, if X_d'' and X_q'' are not equal, then X_2 will not be equal to X_1 . If the difference between X_2 and X_1 of Thevenin's equivalent of the whole system is very high, then it is advised to calculate $\overline{KE}$ from offline simulation for all types of fault.

Observation 2: The $\overline{KE}$ is almost unaffected by a small variation in the fault duration.

Explanation: The accelerating power of generators may vary with fault duration because of change in generator angles. For each generator, output power consists of a) power fed to local loads (shunt admittance) connected at generator buses which is nearly independent of generator angles (completely independent for reduced network internal bus representation), and b) power fed to other load buses and other generators which depends on angular difference among generators.

- 1) For the generators closer to a fault, the drop in the output power and consequently the accelerating power is very high. This drop is mainly due to an increase in the shunt susceptance and the net power fed to the fault is independent of the generator angle.
- 2) Power fed to other load and generator buses is driven by the angle difference between generators. Generators close to each other have a higher equivalent admittance in between compared to electrically distant generators. Therefore, the change in a generator output power is dominated by the change in the angle difference between electrically close generators. Since these generators also accelerate together

during the fault, changes in the fault duration affect the angle difference between these generators much less than the angle difference between distant generators. Therefore, during a fault the accelerating power varies very little.

- 3) Generator angle changes due to change in angular position of rotor which is a relatively slow change. Therefore, with small variations in fault duration, the angle difference among generators closer to each other will not change much. Hence the change in output power and the accelerating power is also very less.

Thus, the accelerating power for generators close to a fault remains almost constant. As the kinetic energy gained by a generator located far away from the fault location is relatively smaller, its ability to affect the Euclidean distance, D_b , is also limited. It can eliminate small error in detecting exact fault inception instant.

Similar approximation were used in transient energy function analysis to find out the system states at fault clearing with constant acceleration approach [35, Sec. 6.4].

REFERENCES

- [1] S. M. Rovnyak, K. Mei, and G. Li, “Fast load shedding for angle stability control,” in *Proc. IEEE Power Eng. Soc. General Meeting*, Jul. 13–17, 2003.
- [2] K. Mei and S. M. Rovnyak, “Response-based decision trees to order stabilizing control,” *IEEE Trans. Power Syst.*, vol. 19, no. 1, pp. 531–537, Feb. 2004.
- [3] G. Andersson *et al.*, “Causes of the 2003 major grid blackouts in North America and Europe, recommended means to improve system dynamic performance,” *IEEE Trans. Power Syst.*, vol. 20, no. 4, pp. 1922–1928, Nov. 2005.
- [4] T. Amraee and S. Ranjbar, “Transient instability prediction using decision tree technique,” *IEEE Trans. Power Syst.*, vol. 28, no. 3, pp. 3028–3037, Aug. 2013.
- [5] I. Kamwa, S. R. Samantaray, and G. Joos, “Development of rule-based classifiers for rapid stability assessment of wide-area post-disturbance records,” *IEEE Trans. Power Syst.*, vol. 24, no. 1, pp. 258–270, Feb. 2009.
- [6] J. Yan, C.-C. Liu, and U. Vaidya, “PMU-based monitoring of rotor angle dynamics,” *IEEE Trans. Power Syst.*, vol. 26, no. 4, pp. 2125–2133, Nov. 2011.
- [7] F. R. Gomez, A. D. Rajapakse, U. D. Annakkage, and I. T. Fernando, “Support vector machine-based algorithm for post-fault transient stability status prediction using synchronized measurements,” *IEEE Trans. Power Syst.*, vol. 26, no. 3, pp. 1474–1483, Aug. 2011.
- [8] F. F. Wu and W. E. Liu, “Detection of topology errors by state estimation,” *IEEE Trans. Power Syst.*, vol. 4, no. 1, pp. 176–183, Feb. 1989.
- [9] Z. Jun and A. Abur, “Identification of network parameter errors,” *IEEE Trans. Power Syst.*, vol. 21, no. 2, pp. 586–592, May 2006.
- [10] I. S. Costa and J. A. Leao, “Identification of topology errors in power system state estimation,” *IEEE Trans. Power Syst.*, vol. 8, no. 4, pp. 1531–1538, Nov. 1993.
- [11] D. Singh, J. Pandey, and D. Chauhan, “Topology identification, bad data processing, and state estimation using fuzzy pattern matching,” *IEEE Trans. Power Syst.*, vol. 20, no. 3, pp. 1570–1579, Aug. 2005.

- [12] N. Singh and H. Glavitsch, "Detection and identification of topological errors in online power system analysis," *IEEE Trans. Power Syst.*, vol. 6, no. 1, pp. 324–331, Feb. 1991.
- [13] M. He and J. Zhang, "A dependency graph approach for fault detection and localization towards secure smart grid," *IEEE Trans. Smart Grid*, vol. 2, no. 2, pp. 342–351, 2011.
- [14] H. Zhu and G. B. Giannakis, "Sparse overcomplete representations for efficient identification of power line outages," *IEEE Trans. Power Syst.*, vol. 27, no. 4, pp. 2215–2224, Nov. 2012.
- [15] J. Wu, J. Xiong, and Y. Shi, "Efficient location identification of multiple line outages with limited PMUs in smart grids," *IEEE Trans. Power Syst.*, to be published.
- [16] J.-C. Chen, W.-T. Li, C.-K. Wen, J.-H. Teng, and P. Ting, "Efficient identification method for power line outages in the smart power grid," *IEEE Trans. Power Syst.*, vol. 29, no. 4, pp. 1788–1800, Jul. 2014.
- [17] T. Banerjee, Y. C. Chen, A. D. Dominguez-Garcia, and V. V. Veeravalli, "Power system line outage detection and identification – A quickest change detection approach," in *Proc. IEEE Conf. Acoustics, Speech, and Signal Processing*, May 2014, pp. 3450–3454.
- [18] A. Y. Abdelaziz, S. F. Mekhamer, M. Ezzat, and E. F. El-Saadany, "Line outage detection using support vector machine (SVM) based on the phasor measurement units (PMUs) technology," in *Proc. IEEE PES General Meeting*, 2012, pp. 1–8.
- [19] J. E. Tate and T. J. Overbye, "Line outage detection using phasor angle measurements," *IEEE Trans. Power Syst.*, vol. 23, no. 4, pp. 1644–1652, Nov. 2008.
- [20] J. E. Tate and T. J. Overbye, "Double line outage detection using phasor angle measurements," in *Proc. IEEE PES General Meeting*, 2009, pp. 1–5.
- [21] J. Wu, J. Xiong, and Y. Shi, "Ambiguity group based location recognition for multiple power line outages in smart grids," in *Proc. IEEE PES Innovative Smart Grid Technologies Conf.*, 2014, pp. 1–5.
- [22] K. Mei, S. M. Rovnyak, and C. M. Ong, "Clustering-based dynamic event location using wide-area phasor measurements," *IEEE Trans. Power Syst.*, vol. 23, no. 2, pp. 673–679, May 2008.
- [23] R. Dashti and J. Sadeh, "Fault section estimation in power distribution network using impedance fault distance calculation and frequency spectrum analysis," *IET Gener., Transm., Distrib.*, vol. 8, no. 8, pp. 1406–1417, 2014.
- [24] C. S. Chen, Y. L. Ke, and J. S. Wu, "Coloured Petri nets approach for solving distribution system contingency by considering customer load patterns," *Proc. Inst. Elect. Eng., Gener., Transm., Distrib.*, vol. 148, no. 5, pp. 463–470, Sep. 2001.
- [25] J. H. Friedman, J. Bentely, and R. A. Finkel, "An algorithm for finding best matches in logarithmic expected time," *ACM Trans. Math. Softw.*, vol. 3, p. 209, 1977.
- [26] P. Kundur, *Power System Stability and Control*. New York, NY, USA: McGraw-Hill, 1994, p. 813.
- [27] M. A. Pai, *Energy Function Analysis for Power System Stability*. Norwell, MA, USA: Kluwer, 1989.
- [28] 246 Bus Reduced NRP G System Data [Online]. Available: http://www.iitk.ac.in/ee/facilities/Research_labs/Power_System/NRPG-DATA.pdf
- [29] P. M. Anderson and A. A. Fouad, *Power System Control and Stability*, 2nd ed. New York, NY, USA: Wiley-IEEE Press, 2002.
- [30] N. Senroy, G. T. Heydt, and V. Vittal, "Decision tree assisted controlled islanding," *IEEE Trans. Power Syst.*, vol. 21, no. 4, pp. 1790–1797, Nov. 2006.
- [31] Y. Li, H.-D. Chiang, B.-K. Choi, Y.-T. Chen, D.-H. Huang, and M. G. Lauby, "Representative static load models for transient stability analysis: Development and examination," *IET Gener., Transm., Distrib.*, vol. 136, no. 2, pp. 68–77, 2007.
- [32] J. Machowski, J. W. Bialek, and J. R. Bumby, *Power System Dynamics: Stability and Control*, 2nd ed. New York, NY, USA: Wiley, 2008.
- [33] P. W. Sauer and M. A. Pai, *Power System Dynamics and Stability*. Englewood Cliffs, NJ, USA: Prentice Hall, 1998.
- [34] S. H. Horowitz and A. G. Phadke, "Third zone revisited," *IEEE Trans. Power Del.*, vol. 21, no. 1, pp. 23–29, Jan. 2006.
- [35] A. A. Fouad and V. Vittal, *Power System Transient Stability Analysis Using the Transient Energy Function Method*. Englewood Cliffs, NJ, USA: Prentice Hall, 1992.

Pratyasa Bhui received the M.Tech degree in power and energy system from the Indian Institute of Technology Kharagapur, Kharagpur, India, in 2013. He is currently pursuing the Ph.D. degree in the Department of Electrical Engineering at IIT Delhi, New Delhi, India.

His research interests include power system dynamics and wide area measurement system.

Nilanjan Senroy (M'06) received the M.S. and Ph.D. degrees from Arizona State University, Tempe, AZ, USA.

He also has postdoctoral experience at the Center for Advanced Power Systems, Florida State University, Tallahassee, FL, USA. He is an Associate Professor in the Department of Electrical Engineering, Indian Institute of Technology, Delhi, India. His research interests include power system stability and control, signal processing, and renewable energy.